

Justin Lee

Staff Software Engineer

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SKILLS

Programming Languages: HTML, CSS, JavaScript, TypeScript, Python, Java, Golang, PHP, C/C++, SQL
Backend: Node.js, Django, Flask, FastAPI, Spring Boot, WebSocket, GraphQL, RESTful APIs, OAuth, JWT
Frontend: React, Next.js, Angular, Redux, Bootstrap, Tailwind CSS, Material UI, D3.js, ApexCharts.js
AI & ML: OpenAI, LLM, Langchain, Embeddings, RAG, Generative AI, Vector Database, LLAMA, GPT
Databases: SQL, MySQL, PostgreSQL, NoSQL, MongoDB, Supabase, AWS DynamoDB, AWS RDS
Cloud services: AWS (EC2, ECS, Bedrock, Lambda), Azure (App Service, Functions, PubSub), GCP
Deployment & CI/CD: Apache, Nginx, Docker, Kubernetes, Git, GitHub, GitLab, GitHub Actions, Jenkins
Methodologies & Practices: Agile, Scrum, Kanban, MVC, A/B Testing, Unit Testing, Jest, Enzyme, CI/CD
Others: Kafka, RabbitMQ, Redis, Google Maps, WebRTC, SEO, Figma, React Query, Google Analytics.

PROFESSIONAL EXPERIENCE

Amazon

Staff Software Engineer

Jul 2019 – Present

- Designed and built **Amazon Haul**, a scalable, **AI-powered, microservices-based** ecommerce platform with a backend using **Java, Spring Boot**, and **AWS**, and a high-performance frontend using **Next.js, TypeScript**, and **Tailwind CSS**, delivering dynamic product pages, personalized home feeds, and adaptive carts that improved user engagement and significantly increased mobile conversion rates.
- Built an **AI-powered recommendation system** using **Python / FastAPI** and **Amazon's LLM ecosystem**, integrating user behavior signals and product metadata to deliver personalized suggestions across the shopping journey, resulting in a measurable uplift in customer satisfaction and a 25% increase in sales.
- Led a small team of 5+ engineers in designing and developing the AI prototype for **Amazon's Rufus Agents**, leveraging **Python / FastAPI** and **RAG** pipeline to provide conversational shopping assistance, significantly enhancing user experience and driving a 30% increase in customer engagement.
- Built **RESTful APIs** with **Spring Boot** for the primary backend of **Amazon Haul**, implementing secure authentication and authorization using **OAuth 2.0, JWT** and **AWS IAM policies** to protect sensitive user data and securely ensure **PCI-compliant** ecommerce transactions.
- Built a modular, component-driven frontend for the ecommerce platform with **Next.js, TypeScript**, and **Tailwind CSS**, introducing **micro-frontends** to enable independent deployments, cut merge conflicts by 60%.

- Integrated **AWS Graviton-based EC2 and Lambda instances** to reduce latency and cost, optimizing key backend APIs such as checkout, product search, and recommendations, achieving up to 20% faster response times.
- Connected **third-party logistics APIs** such as USPS and UPS for real-time order tracking and delivery updates, improving shipment visibility and reducing customer support inquiries by 40%.
- Automated frontend and backend testing using **Jest, Cypress** and **JUnit**, achieving over 95% coverage, and streamlined **CI/CD** pipelines with **GitHub Actions** and **Jenkins** to reduce deployment time and improve reliability.
- Leveraged Agile ceremonies including sprint planning, daily standups, and retrospectives; collaborated with product managers to refine requirements, mentored junior engineers through code reviews and pair programming, and enforced engineering best practices to maintain high code quality and velocity.

Amwell

Senior Software Engineer

Jun 2015 – Jul 2019

- Contributed significantly to the migration of legacy telehealth services to the new cloud-native Converge platform, refactoring monolithic **Java / Spring Boot** applications into microservices on **AWS (ECS, Lambda, RDS)**, reducing system downtime and increasing virtual visit success rates by over 20%.
- Developed **end-to-end** video consultation features using **WebRTC** and secure SIP/WebSocket signaling, integrated with EHR systems (Cerner, Epic) via HL7/FHIR interfaces to streamline scheduling and visit workflows.
- Built **RESTful APIs** with **Java / Spring Boot** for the primary backend of the video consultation system, ensuring reliable communication between services and implementing robust data protection practices to support HIPAA-compliant handling of PHI in a cloud-native telehealth environment.
- Architected and developed modular, scalable frontends for telehealth services using **React** and **micro-frontend** architecture, enabling independent deployment of **video consultations** and **scheduling modules**.
- Developed monitoring scripts and event-driven alerts with **AWS CloudWatch** and **Lambda** to support system health tracking and improve uptime to 99.9%.
- Utilized **Apache Kafka** as a distributed **message queue system** to decouple backend services and improve scalability and throughput, supporting thousands of concurrent sessions with increased fault tolerance.
- Designed and optimized relational database schemas, implementing **database indexing, query tuning, partitioning**, and replication to enhance performance and scalability while ensuring HIPAA-compliant data integrity and access control.
- Supported containerization efforts using **Docker** and **Kubernetes** and participated in code reviews and knowledge-sharing sessions to help modernize the team's DevOps culture.

EDUCATION

San Francisco State University

Aug 2011 – May 2015

Bachelor's degree in computer science